

Linux Command Cheat Sheet

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Linux Command Cheat Sheet

File and Directory Operations

Navigation

- `pwd` - Print working directory (shows current location)
- `cd /path/to/directory` - Change directory
- `cd ..` - Move up one directory level
- `cd ~` or `cd` - Go to home directory
- `cd -` - Go to previous directory

Listing Files

- `ls` - List files and directories
- `ls -l` - Long format (detailed view with permissions, owner, size, date)
- `ls -la` - Long format including hidden files (starting with .)
- `ls -lh` - Long format with human-readable file sizes
- `ls -lt` - Sort by modification time (newest first)
- `ls -ltr` - Sort by modification time (oldest first)

Creating Files and Directories

- `touch filename` - Create empty file or update timestamp
- `mkdir dirname` - Create directory
- `mkdir -p path/to/nested/dirs` - Create nested directories
- `mkdir -m 755 dirname` - Create directory with specific permissions

Copying, Moving, and Deleting

- `cp source dest` - Copy file
- `cp -r source_dir dest_dir` - Copy directory recursively
- `cp -i source dest` - Interactive copy (prompt before overwrite)
- `mv source dest` - Move/rename file or directory
- `rm filename` - Delete file
- `rm -r dirname` - Delete directory recursively
- `rm -rf dirname` - Force delete directory (use with caution!)
- `rm -i filename` - Interactive delete (prompt before deletion)

Finding Files

- `find /path -name "filename"` - Find files by name
- `find /path -type f -name "*.txt"` - Find files with specific extension
- `find /path -type d -name "dirname"` - Find directories by name
- `find /path -size +100M` - Find files larger than 100MB
- `find /path -mtime -7` - Find files modified in last 7 days
- `locate filename` - Quick file search using database

- `which command` - Find location of executable
- `whereis command` - Find binaries, source, and manual pages

File Content Operations

Viewing Files

- `cat filename` - Display entire file content
- `less filename` - View file page by page (q to quit)
- `more filename` - View file page by page (older version of less)
- `head filename` - Show first 10 lines
- `head -n 20 filename` - Show first 20 lines
- `tail filename` - Show last 10 lines
- `tail -n 20 filename` - Show last 20 lines
- `tail -f filename` - Follow file changes (real-time monitoring)

Editing Files

- `nano filename` - Simple text editor
- `vim filename` - Advanced text editor
- `emacs filename` - Another advanced text editor

Searching in Files

- `grep "pattern" filename` - Search for pattern in file
- `grep -i "pattern" filename` - Case-insensitive search
- `grep -r "pattern" directory` - Recursive search in directory
- `grep -n "pattern" filename` - Show line numbers
- `grep -v "pattern" filename` - Show lines NOT containing pattern
- `grep -c "pattern" filename` - Count matching lines

Text Processing

- `sort filename` - Sort lines alphabetically
- `sort -n filename` - Sort numerically
- `sort -r filename` - Reverse sort
- `uniq filename` - Remove duplicate lines (requires sorted input)
- `wc filename` - Count lines, words, and characters
- `wc -l filename` - Count lines only
- `cut -d',' -f1,3 filename` - Extract columns from CSV
- `awk '{print $1}' filename` - Print first column
- `sed 's/old/new/g' filename` - Replace text globally

File Permissions and Ownership

Understanding Permissions

```
-rwxrwxrwx
- = file type (- for file, d for directory)
rwx = owner permissions (read, write, execute)
rwx = group permissions
rwx = other permissions
```

Changing Permissions

- `chmod 755 filename` - Set permissions using octal notation
- `chmod u+x filename` - Add execute permission for owner
- `chmod g-w filename` - Remove write permission for group
- `chmod o+r filename` - Add read permission for others
- `chmod a+r filename` - Add read permission for all

Octal Permission Values

- 4 = read (r)
- 2 = write (w)
- 1 = execute (x)
- 7 = read + write + execute (4+2+1)
- 6 = read + write (4+2)
- 5 = read + execute (4+1)

Changing Ownership

- `chown user filename` - Change file owner
- `chown user:group filename` - Change owner and group
- `chgrp group filename` - Change group only
- `chown -R user:group directory` - Change ownership recursively

Process Management

Viewing Processes

- `ps` - Show running processes for current user
- `ps aux` - Show all running processes
- `ps -ef` - Show all processes with full format
- `top` - Real-time process viewer
- `htop` - Enhanced interactive process viewer
- `pgrep process_name` - Find process ID by name
- `pidof process_name` - Get process ID

Controlling Processes

- `kill PID` - Terminate process by ID
- `kill -9 PID` - Force kill process
- `killall process_name` - Kill all processes by name
- `nohup command &` - Run command in background, ignore hangup

- `command &` - Run command in background
- `jobs` - List background jobs
- `fg %1` - Bring job 1 to foreground
- `bg %1` - Send job 1 to background

System Information

System Status

- `uname -a` - System information
- `whoami` - Current username
- `id` - User and group IDs
- `date` - Current date and time
- `uptime` - System uptime and load
- `w` - Who is logged in and what they're doing
- `last` - Last logged in users

Hardware Information

- `lscpu` - CPU information
- `lsmem` - Memory information
- `lsblk` - Block devices (disks)
- `lsusb` - USB devices
- `lspci` - PCI devices
- `dmidecode` - Hardware information from BIOS

Disk Usage

- `df -h` - Disk space usage (human-readable)
- `du -h directory` - Directory size
- `du -sh directory` - Total size of directory
- `du -h --max-depth=1` - Size of immediate subdirectories
- `free -h` - Memory usage

Network Commands

Network Information

- `ip addr` - Show IP addresses
- `ifconfig` - Network interface configuration (older)
- `ping hostname` - Test connectivity
- `ping -c 4 hostname` - Ping 4 times only
- `traceroute hostname` - Trace network path
- `netstat -tuln` - Show listening ports
- `ss -tuln` - Modern replacement for netstat

Network Tools

- `wget url` - Download file from web

- `curl url` - Transfer data from/to server
- `ssh user@hostname` - Secure shell connection
- `scp file user@host:/path` - Secure copy over network
- `rsync -av source dest` - Sync files/directories

Archive and Compression

Creating Archives

- `tar -cvf archive.tar files` - Create tar archive
- `tar -czvf archive.tar.gz files` - Create compressed tar archive
- `zip archive.zip files` - Create zip archive

Extracting Archives

- `tar -xvf archive.tar` - Extract tar archive
- `tar -xzvf archive.tar.gz` - Extract compressed tar archive
- `unzip archive.zip` - Extract zip archive

Archive Options

- `c` = create
- `x` = extract
- `v` = verbose
- `f` = file
- `z` = gzip compression
- `j` = bzip2 compression

Input/Output Redirection

Redirection Operators

- `command > file` - Redirect output to file (overwrite)
- `command >> file` - Append output to file
- `command < file` - Use file as input
- `command 2> file` - Redirect error output to file
- `command &> file` - Redirect both output and errors

Pipes

- `command1 | command2` - Pipe output of command1 to command2
- `command | grep pattern` - Filter command output
- `command | sort` - Sort command output
- `command | head -10` - Show first 10 lines of output

Environment and Variables

Environment Variables

- `echo $HOME` - Display home directory

- `echo $PATH` - Display executable path
- `env` - Show all environment variables
- `export VAR=value` - Set environment variable
- `unset VAR` - Remove environment variable

Command History

- `history` - Show command history
- `!!` - Repeat last command
- `!n` - Repeat command number n
- `!string` - Repeat last command starting with string
- `ctrl+r` - Search command history interactively

Text Editors Quick Reference

Nano

- `ctrl+o` - Save file
- `ctrl+x` - Exit
- `ctrl+w` - Search
- `ctrl+k` - Cut line
- `ctrl+u` - Paste

Vim Basics

- `i` - Insert mode
- `esc` - Normal mode
- `:w` - Save
- `:q` - Quit
- `:wq` - Save and quit
- `:q!` - Quit without saving

Useful Keyboard Shortcuts

Command Line

- `ctrl+c` - Cancel current command
- `ctrl+z` - Suspend current command
- `ctrl+d` - End of file/logout
- `ctrl+l` - Clear screen
- `ctrl+a` - Move to beginning of line
- `ctrl+e` - Move to end of line
- `ctrl+u` - Delete from cursor to beginning
- `ctrl+k` - Delete from cursor to end
- `tab` - Auto-complete
- `tab tab` - Show all possible completions

Package Management (Debian/Ubuntu)

APT Commands

- `apt update` - Update package list
- `apt upgrade` - Upgrade installed packages
- `apt install package` - Install package
- `apt remove package` - Remove package
- `apt search package` - Search for package
- `apt show package` - Show package information

Helpful Tips

Wildcards

- `*` - Match any characters
- `?` - Match single character
- `[abc]` - Match any character in brackets
- `[a-z]` - Match any character in range

Common File Locations

- `/etc` - System configuration files
- `/var/log` - Log files
- `/home` - User home directories
- `/tmp` - Temporary files
- `/usr/bin` - User binaries
- `/usr/local/bin` - Local binaries

Getting Help

- `man command` - Manual page for command
- `command --help` - Quick help for command
- `info command` - Info page for command
- `apropos keyword` - Find commands related to keyword